



Evolution of the PtPdCuNiCo high-entropy alloy nanoparticles structure during thermal treatment: Modeling and in situ study

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ABSTRACT

This manuscript reports the first systematic study of the structural evolution of the high-entropy alloy (HEA) PtPdCuNiCo/C catalyst composition during heat treatment, obtained via a simple one-pot borohydride synthesis. *In situ* X-ray diffraction data revealed phase separation of the alloy upon heating at 400 °C and above. A more detailed study of the structure of the materials before and after phase separation revealed that heat treatment partially reduced oxidized metals to the metallic phase. Calculation of the surface composition of high-entropy alloy nanoparticles based on the results from hybrid MD/MC simulations shows a significant enrichment of the nanoparticle surface with platinum, which correlates well with the electrochemical behavior of the catalysts. Heat treatment leads, on the one hand, to platinum enrichment of the surface and an increase in specific surface activity, and, on the other hand, to nanoparticle coarsening and a decrease in electrochemically active surface area. Under the conditions of this study, heat treatment at 350 °C strikes a balance between enhanced activity and avoidance of significant phase separation. The excellent stability of the PtPdCuNiCo/C catalyst after heat treatment at 350 °C has been demonstrated; the activity in the oxygen reduction reaction of the HEA catalyst after stress testing is four times higher than that of the Pt/C benchmark. Understanding the structural evolution of complex multicomponent nanoparticles is important for the targeted selection of optimal post-treatment conditions to create high-performance catalysts for low-temperature fuel cells.

1. Introduction

The development of modern energy technologies, including low-temperature fuel cells and electrolyzers, requires the creation of a new generation of electrocatalysts that combine high activity, stability, and cost efficiency [1–4]. Previously, the main focus of researchers was on bi- and trimetallic systems based on platinum group metals (platinum, palladium, etc.), which made it possible to achieve a significant improvement in functional characteristics in key reactions such as oxygen reduction, hydrogen oxidation, and oxygen evolution [5,6]. However, the potential for further optimization of such systems by increasing the number of components remains underexplored [7–9].

Of scientific and practical interest is the creation of high-entropy alloys (HEAs) based on platinum group metals (Pt, Pd, Ir, Ru) in

combination with D-metals (Co, Ni, Cu, Fe, Cr) [7,9–15]. Such materials are promising for a wide range of electrochemical processes in both cathodic (oxygen reduction) and anodic (hydrogen oxidation, oxidation of small organic molecules) reactions of fuel cells [16].

Currently, Pt-based HEAs represent the most studied class of oxygen reduction reaction catalysts, typically synthesized by combining platinum with other metals that have similar lattice parameters and crystal structures to ensure miscibility [17]. The PtPdCoNiCu/C catalyst was selected for the study, as Co, Ni, and Cu are the elements most commonly used for doping platinum, due to the high activity of Pt-based catalysts with these elements [18–23]. It has been demonstrated in a number of studies that alloying platinum with palladium enhances the ORR activity relative to monometallic Pt/C catalysts [24–27]. Given that global palladium reserves slightly exceed those of platinum, the adoption of

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Pt_xPd/C systems could enhance the accessibility of electrocatalysts for the expanding hydrogen technology market [26]. It is expected that this choice of elements will promote the formation of a single-phase alloy with a stable face-centered cubic (FCC) structure.

One of the key factors determining the functional characteristics of HEA-based catalysts is the synthesis method, as well as the subsequent treatment of such materials (thermal or acid treatment). The synthesis method directly affects the nanoparticle size, their microstructure, and morphology, which collectively determine the number and nature of active sites, and consequently, the features of their electrochemical behavior [7,13,28–31]. Post-treatment of such catalysts is also an important stage, largely determining their final structural and functional characteristics. Thus, thermal treatment plays a key role in completing the process of further alloying of the metallic components due to atomic interdiffusion [32–36]. In addition, in some cases, the wet synthesis of HEAs leads to the formation of a mixture of metals and their oxides/hydroxides [37], which necessitates additional heat treatment in a reducing atmosphere. An optimally selected temperature regime makes it possible not only to eliminate component segregation but also to form a surface composition that ensures high catalytic activity [33–36,38,39]. A temperature that is too low will not ensure complete alloying of the elements, while a temperature that is too high will lead to the aggregation and sintering of nanoparticles, which reduces the specific surface area of the catalyst. Furthermore, the difficulty of applying heat treatment to HEA-based catalysts is determined by the lack of understanding and predictability of the behavior of such systems both during their synthesis and during their post-treatment. It is important to emphasize that heat treatment is often used for the synthesis of catalysts based on HEAs [40,41], while the post-treatment of already prepared catalysts has been studied insufficiently.

The design of efficient heterogeneous catalysts based on HEA nanoparticles requires a fundamental understanding of their surface structure and composition under operating conditions. It is now well established that the surface of a multi-metallic particle is not merely a random slice of the bulk. Instead, it is subject to surface segregation, a thermodynamic phenomenon in which elements enrich the surface layer to minimize the overall free energy of the system [42,43]. The driving forces for this process are complex and arise from a combination of factors, including differences in elemental surface energies, atomic sizes, and the strength of interactions with adsorbates present in the environment. Consequently, accurately predicting the equilibrium surface structure of a HEA nanoparticle is a critical prerequisite for reliably assessing its catalytic activity [44].

Early theoretical efforts to understand segregation in HEAs laid the groundwork by combining first-principles calculations with simplified thermodynamic models. Ferrari and Körmann [45], for instance, used density functional theory (DFT) to investigate the prototypical Cr-Mn-Fe-Co-Ni (Cantor) alloy. Their work demonstrated that in vacuum, segregation is relatively weak, with a slight tendency for Ni to surface-segregate in Cr-, Mn-, and Fe-rich alloys. However, they predicted a drastic change in the presence of an oxygen adsorbate, leading to a very strong surface enrichment of Cr and Mn across all compositions. This study was pivotal in showing that the reactive environment can completely invert segregation trends, a conclusion rationalized through canonical *d*-band models [45].

Following this, Kristoffersen and Rossmeisl [46] extended the investigation to another important quinary system, Ag-Au-Cu-Pd-Pt. By constructing large ensembles of random structures and approximating the canonical partition function, they were able to model local order and segregation at finite temperatures. Their work revealed that the type and degree of surface order are strongly facet-dependent. For the (111) surface, they observed significant vertical segregation (enrichment of Au and Ag in the topmost layer), but within that layer the lateral distribution of nearest-neighbor atoms remains nearly random – meaning that while the composition of the surface differs strongly from the bulk, the atoms are not arranged in an ordered pattern relative to each other. In

contrast, the (100) surface exhibits both strong segregation and a highly ordered distribution of nearest-neighbor pairs, indicating the formation of a specific local atomic arrangement. Crucially, they proposed that the width of the structure ensemble's energy distribution could serve as a computationally inexpensive proxy for estimating the degree of local order in an alloy [46,47].

While these seminal studies provided invaluable atomistic insights, they were primarily confined to periodic slab models of planar surfaces. Real-world catalysts, however, exist as nanoparticles, where under-coordinated atoms at corners and edges, as well as finite-size effects, play a dominant role. The immense computational cost of DFT prohibits its direct application to the large, multi-thousand atom nanoparticles typical of experimental catalysts (e.g., 3–7 nm in diameter) across the vast HEA compositional space. To bridge this gap, more recent works have turned to machine learning. Cao and co-workers, for example, developed a DFT-trained machine-learning cluster expansion (ML-CE) framework to perform high-throughput screening of quinary Ir-Pd-Pt-Rh-Ru nanoparticles. Their approach, coupled with Monte Carlo simulations, successfully resolved temperature- and atmosphere-dependent surface segregation and predicted compositions with significantly enhanced activity for the oxygen reduction reaction (ORR), linking performance to specific local coordination motifs like first-nearest-neighbor triplets [44]. This demonstrated the power of surrogate models to navigate the immense configurational space of HEA nanoparticles and make experimentally verifiable predictions.

Thus, the aim of this study is to experimentally establish the relationship between the conditions of thermal post-treatment, microstructure, surface composition, and electrochemical characteristics of high-entropy PtPdCoNiCu/C electrocatalysts, as well as to perform hybrid MD/MC modeling of the structural evolution of HEA nanoparticles upon heating. *In situ* XRD analysis of the as-prepared catalyst revealed that heating at ≥ 400 °C induces phase separation. It was found that 350 °C is optimal, providing surface platinum enrichment, increased specific activity, and preserved electrochemically active surface area. After stress testing, the catalyst treated at 350 °C shows fourfold higher ORR activity than the Pt/C benchmark.

2. Materials and methods

The PtPdCuNiCo/C catalysts were synthesized via simultaneous reduction of the corresponding metal precursors (H₂PtCl₆, PdCl₂, CuSO₄·5H₂O, NiSO₄·7H₂O, CoSO₄·7H₂O) in an aqueous–ethylene glycol suspension (water/ethylene glycol volume ratio = 1/3) of Vulcan XC–72 carbon support, using sodium tetrahydroborate as the reducing agent [48,49]. The suspension was prepared by mixing the carbon carrier with a metal precursors solution on a magnetic stirrer, followed by ultrasonic homogenization. After dispersion, a freshly prepared 0.5 mol/L aqueous solution of sodium tetrahydroborate was added dropwise in a tenfold molar excess under constant stirring of the suspension on a magnetic stirrer at room temperature. After the addition, the suspension was left stirring for one hour. The resulting material was filtered using a Büchner funnel with a "Blue Ribbon" paper filter, washed with bidistilled water, diluted isopropyl alcohol (alcohol/water = 1/2), and again with bidistilled water. The obtained catalyst was then dried in a vacuum drying oven at 70 °C.

Thermal treatment was carried out in a tubular furnace using two different protocols: at 350 °C under an argon atmosphere, and at 600 °C under an argon atmosphere with the addition of 5% H₂. The obtained samples were designated as BG, BG-350, and BG-600 for the catalyst prepared by the borohydride synthesis method, after thermal treatment at 350 °C and 600 °C, respectively.

The surface chemical composition of the samples was studied by X-ray photoelectron spectroscopy (XPS) using a custom-made X-ray photoelectron spectrometer equipped with an XR–50 X-ray source with a double Al/Mg anode and a PHOIBOS–150 hemispherical electron energy analyzer with a 9-channel MCD detector (SPECS Surface Nano

Analysis GmbH, Germany). The spectrometer is equipped with a special high-pressure cell, allowing sample treatment in various environments and their transfer to the analyzer chamber without air exposure (*pseudo in situ* XPS mode). The design of the spectrometer and the experimental methodology are described in detail elsewhere [50]. Powders of the studied catalysts were pressed into a metal net, which was fixed on a sample holder. Non-monochromatized Al K α radiation ($h\nu = 1486.6$ eV) was used to record the core-level spectra. The charge correction was performed by setting the C1s peak at 284.5 eV. Data processing was performed using the CasaXPS software. After background subtraction using the Shirley method, the experimental curve was approximated by a series of peaks corresponding to electron photoemission from atoms in different chemical environments. The peaks were fitted using a Gaussian and Lorentzian blend for each component. The peak asymmetry in the Pt4f, Pd3d, Cu2p, and Co2p spectra was defined in the form of a tail-dampened Lorentzian asymmetric line shape given in CasaXPS by LF (α, β, w, m), where α and β define the spread of the tail on either side of the Lorentzian component, m specifies the width of the Gaussian used to convolute the Lorentzian curve and w is the tail dampening parameter. Relative concentrations of elements in the analysis area were determined based on the integrated intensities of the XPS peaks. XPS spectra were recorded before and after the treatment in the high-pressure cell for 60 min in an atmosphere of 5% H₂/Ar at 350 °C.

In situ X-ray diffraction (XRD) experiments were carried out using a Bruker D8 Advance diffractometer (Bruker, Germany) equipped with an XRD-900 reactor chamber (Anton Paar GmbH, Austria). The diffraction patterns were obtained using Cu K α radiation ($\lambda = 1.5418$ Å) in the 2 θ angle range of 15–95° with a step of 0.05°. The sample was heated to the specified temperatures (300, 400, 500, and 600 °C) in a reducing atmosphere (5% H₂ in He). XRD patterns of the samples were also obtained in their initial state and after treatment at room temperature. Rietveld analysis was performed using X'Pert HighScore Plus software. The crystallite size values were calculated using the Scherrer equation.

High-resolution transmission electron microscopy (TEM) studies were carried out using a Themis Z microscope equipped with a high-performance Ceta 16 M CMOS-based matrix detector and a Super-X energy-dispersive X-ray spectroscopy detector (Thermo Fisher Scientific, Inc., USA). The microscopy was operated at an accelerating voltage of 200 kV. The first detector was used to obtain high angle annular dark-field (HAADF) images in the scanning mode (STEM) with a lattice resolution of 0.07 nm; the second detector was used for energy-dispersive X-ray spectroscopy (EDX) and EDX mapping. The samples for study were prepared on aluminum grids using ultrasonic dispersion of the catalysts in ethanol for the standard experiments.

Thermal analysis of the obtained materials was performed using a combined thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) analyzer (NETZSCH STA 449 C Jupiter). The measurements were conducted in an atmosphere of 80% N₂ and 20% O₂, within a temperature range from 40 to 800 °C, at a heating rate of 10 °C/min and a gas flow rate of 20 ml/min, using alumina crucibles.

Electrochemical measurements employed a standard three-electrode system using an Ag/AgCl (3.5 M KCl) reference electrode and platinum mesh counter electrode in 0.1 M HClO₄ and 0.1 M KOH electrolytes. All potentials are referenced to the reversible hydrogen electrode (RHE). Catalyst inks were prepared by ultrasonically dispersing 4.0 mg catalyst with 50 μ L deionized water, 1.95 ml isopropyl alcohol, and 21 μ L 5 wt% Nafion® solution for 20 min, then drop-casting onto a glassy carbon RDE (5 mm diameter) to achieve a 20.0–22.0 μ gPt/cm² loading. Electrodes were activated via 100 CV cycles (0.04–1.00 V, 200 mV/s) followed by 2 cycles (20 mV/s) in Ar-saturated electrolyte. The electrochemically active surface area (ECSA) was determined from hydrogen adsorption/desorption charges (0.04–0.40 V) using 210 μ C/cm²Pt as the conversion factor. The electrochemical CO oxidation (CO-stripping) was studied by purging the 0.1 M HClO₄ deaerated solution with the CO gas for 10 min, with the working electrode held at $E = 0.1$ V. The cell was then purged with Ar for 30 min to remove excess CO. Next, potential was scanned in

the anodic direction up to 1.10 V at a sweep rate of 20 mV/s. ORR activity was evaluated through linear sweep voltammetry (0.10–1.10 V, 20 mV/s) at 400–2500 rpm in O₂-saturated electrolyte, with kinetic currents extracted via Koutecký-Levich analysis and activities reported as both mass- (A/g(PGM)) and ECSA-normalized (A/m²(PGM)) values at 0.90 V vs. RHE, for measurements performed in both 0.1 M HClO₄ and 0.1 M KOH. The electrochemical characteristics of the obtained materials were compared with those of commercial Pt/C analogs with platinum mass fractions of 20% (JM20, Johnson Matthey) and 40% (JM40, Johnson Matthey).

To evaluate catalyst stability, accelerated stress tests (AST) were performed by applying 10,000 square-wave potential steps between 0.4 and 1.0 V vs. RHE (3 s per step) in an O₂-saturated 0.1 M HClO₄ solution at 25 °C. After the AST protocol, the electrode surface was conditioned by 100 voltammetric cycles in the range of 0.04–1.00 V at 200 mV/s. The post-stress ECSA was determined from two consecutive cyclic voltammograms, and the ORR activity was subsequently assessed via linear sweep voltammetry.

We employ a hybrid molecular dynamics (MD) and Monte Carlo (MC) approach accelerated by a Deep Potential (DeepMD) machine learning force field. This methodology allows us to overcome the time-scale limitations of conventional MD, enabling the simulation of atomic diffusion and elemental swaps necessary to reach a thermodynamically equilibrated state. The system was considered to have reached a thermodynamically equilibrated state when both (i) the fluctuation of the total potential energy decreased below 0.05 eV/atom over 100 ps, and (ii) the layer-by-layer elemental composition varied by less than 2 at% over the same interval. Specifically, we investigate PtPdCuNiCo HEA nanoparticles with a diameter of ~ 3.7 nm (containing 2406 atoms), a size deliberately chosen to match the average size of catalytically relevant nanoparticles synthesized experimentally. Starting from a random solid solution, we simulate the annealing process at temperatures of 50 K, 300 K, 600 K and 900 K to observe the temperature-dependent evolution of the surface composition. By directly tracking the layer-by-layer elemental distribution, we can quantify the equilibrium segregation profile for this multi-principal element system under conditions relevant to catalysis and synthesis.

3. Results and discussion

3.1. *In situ* X-ray diffraction study of the PtPdCuNiCo/C catalyst structural evolution during heat treatment

The PtPdCuNiCo/C catalyst BG, synthesized by the borohydride method, has a composition Pt₁₉Pd₁₆Cu₂₂Ni₂₂Co₂₁ according to X-ray fluorescence analysis, while thermogravimetry shows a metal content of 36.7 wt% (Fig. S1). The calculated platinum group metal (PGM) content, based on this composition, was 21.2 wt%, which is close to the expected value from the precursor loading.

The diffraction pattern of the PtPdCuNiCo/C BG catalyst (Fig. 1) is characterized by a broad peak in the 20–30° region, which corresponds to the carbon support, and broad maxima at $2\theta = 39.9, 46.7, 68.8, 82.6,$ and 87.3° , corresponding to the (111), (200), (220), (311), and (222) reflections of a phase with a cubic lattice and space group Fm-3m. This phase is likely a mixed compound (Pt,Pd,Cu,Co,Ni) containing Pt, Pd, Cu, Co, and Ni in an unknown ratio. The lattice parameter of this phase is 3.864(1) Å, which is somewhat higher than the theoretically calculated value according to Vegard's law for a material with an equal atomic fraction of all components, Pt₂₀Pd₂₀Cu₂₀Ni₂₀Co₂₀, which is 3.701 Å. Thus, the lattice parameter of the metallic phase in the PtPdCuNiCo/C BG catalyst is closer to the parameters corresponding to Pd (3.878 Å, ICSD No. 672508) or Pt (3.923 Å, ICSD No. 243678) than to a solid solution of PtPdCuNiCo, indicating incomplete alloying of the components. Calculation of the crystallite size using the Scherrer equation for the BG material showed a crystallite size of about 3 nm. Also, weak and broad maxima are observed on the diffraction pattern of the BG sample

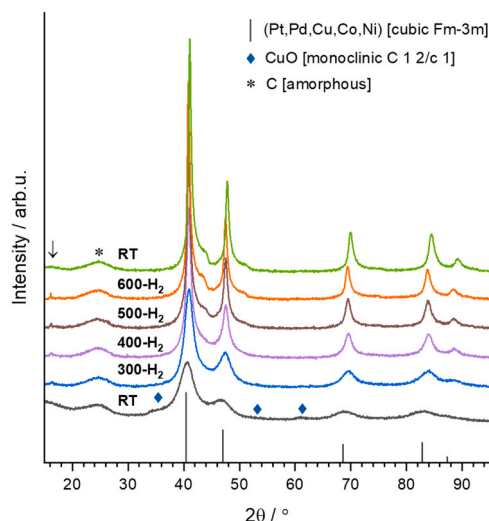


Fig. 1. Diffraction patterns of the PtPdCuNiCo/C BG sample recorded during heating in an H_2/He atmosphere. The arrow indicates an additional peak of an impurity phase.

at approximately 35, 54, and 61°. Due to the significant width of these peaks and their overlap with those of the (Pt,Pd,Cu,Co,Ni) phase, it is difficult to determine to which phase they belong; they possibly correspond to CuO (space group C12/c1, ICSD No. 87125). Additionally, Cu, Co, and Ni that are not incorporated into the alloy may exist as X-ray amorphous oxides.

To increase the degree of alloying of the metallic components and to reduce oxidized metals, heat treatment is performed in an inert atmosphere with hydrogen addition. However, selecting the optimal treatment temperature for each specific catalyst composition and structure is a complex and labor-intensive task. To address this issue, *in situ* methods, particularly *in situ* XRD, are used. These methods make it possible to study changes in the catalyst structure during heating to various temperatures in a single experiment and, based on the obtained data, to select the optimal heat-treatment conditions.

The *in situ* XRD results for the BG material in an H_2/He atmosphere (Fig. 1) show that upon heating to 300 °C, the peaks of the (Pt,Pd,Cu,Co,Ni) phase narrow, and the peaks corresponding to the CuO phase disappear. The lattice parameter decreases to 3.819(1) Å. At 400 °C, segregation of a (Pt,Pd,Cu,Co,Ni)* phase with the same symmetry group but a smaller lattice parameter is observed, manifesting as asymmetry of the peaks on the higher-angle side. The lattice parameter of the (Pt,Pd,Cu,Co,Ni) phase increases to 3.824 Å, while that of the new (Pt,Pd,Cu,Co,Ni)* phase is 3.807(1) Å. In addition, a small diffraction maximum appears at 16°; its nature could not be determined. Notably, this maximum is observed at high temperatures in the diffraction patterns of all samples.

At 500 °C, a (Pt,Pd,Cu,Co,Ni)** phase with narrower peaks at higher angles is formed. This phase has a lattice parameter of 3.574(1) Å. Notably, in this case the diffraction peaks are not sufficiently resolved to unambiguously determine the number of distinct phases; however, the best Rietveld refinement fit was obtained for this case. An example of refinement with peaks from three phases is shown in Fig. S2. At 600 °C, a noticeable narrowing of the diffraction peaks occurs. Upon cooling to room temperature, the maximum at 16° is no longer observed. The lattice parameters of the formed phases are 3.804(1) and 3.753(1) Å, which are significantly lower than the initial phase parameter of 3.864 (1) Å. The change in lattice parameter with temperature is shown in Fig. S3.

Phase separation in the studied systems can occur either via the spinodal decomposition mechanism [51] or via nucleation and growth [52], the latter requiring overcoming an energy barrier. Determining the

specific mechanism would require additional thermal measurements and TEM analysis, which are beyond the scope of this study. Under reducing conditions, segregation is primarily driven by the tendency to minimize surface energy [44]. The outer layer becomes enriched in Pt, which has relatively low surface energy, and this can significantly enhance the activity toward the ORR [44].

For nanoparticles, the size effect becomes a critical factor, significantly shifting phase boundaries compared to bulk materials. Owing to the excess surface energy and confined geometry, the classical signatures of phase separation are modified. Since the melting temperatures of all constituents are considerably higher than 400 °C, it is most likely that the process in question is solid-state phase separation [53]. The elements have different melting temperatures and enthalpies of mixing. At the microscale, during slow cooling, the alloy may tend towards nanoscale phase separation into regions enriched in noble metals (Pt, Pd) and transition metals (Ni, Co, Cu). The presence of copper in the alloy composition can have a significant effect; for instance, nanoscale phase separation due to Cu segregation has been observed in the article by Xu [54]. In future work, we plan to obtain analogous systems in which copper atoms are replaced by Fe, Sn, or Cr, and to conduct similar studies on the thermal stability of the alloys upon heat treatment.

Thus, heating the PtPdCuNiCo/C BG catalyst leads not only to an increase in crystallite size (Table S1) and a decrease in the lattice parameter of the metallic phase, but also to phase separation at temperatures of 400 °C and above. Given these results, to avoid phase separation, it is not recommended to treat this material at or above 400 °C.

3.2. Structural evolution of the PtPdCuNiCo/C catalyst upon heat treatment

To study the effect of treatment temperature on the microstructure and catalytic activity of the materials, *ex situ* treatment of the BG catalyst was performed at temperatures below and above the phase separation temperature: 350 °C and 600 °C, respectively. The diffraction patterns of the obtained catalysts are shown in Fig. S4.

Transmission electron microscopy confirms the formation of metallic nanoparticles uniformly distributed on the carbon support (Figs. 2a and 2b) with an average size of 3.8 nm (Fig. 2d). EDX-TEM compositional mapping of individual areas of the BG catalyst showed the presence of all five metals in the nanoparticles (Figs. 2c and S5), confirming the formation of five-component PtPdCuNiCo metallic nanoparticles.

Heat treatment of the material at 350 °C leads to an increase in the average nanoparticle size by about 1 nm to 4.9 nm, while the width of the size distribution also increases (Fig. 2h), and an additional "tail" appears on the histogram in the 9–11 nm region, indicating the formation of a noticeable fraction of nanoparticle agglomerates after heat treatment [55]. EDX-TEM compositional mapping of individual areas of the BG-350 catalyst, similarly to the BG material, showed the presence of all five metals in the nanoparticles (Fig. 2g and S6), confirming the preservation of the five-component PtPdCuNiCo metallic nanoparticles after heat treatment. Notably, in the HRTEM images of nanoparticles of the BG-350 material, a lighter amorphous shell approximately 1.5–2.0 nm thick and a well-crystallized darker core can be distinguished (Fig. S7a). This may be related to the formation of an oxide film on the nanoparticle surface and requires more detailed study. EDX analysis of the composition of individual nanoparticles in their core and shell regions (Fig. S7c) confirmed the assumption regarding the formation of an oxide film, since the oxygen content in the shell region significantly exceeded that in the nanoparticle core. Analysis of interatomic distances for individual particles revealed a wide range of values, which are nevertheless smaller than the interatomic distances for Pt or Pd, suggesting the formation of multicomponent nanoparticles of various compositions (Fig. S7b).

TGA/DSC of metal-carbon materials enables determination of the total metal content from the residue after carbon support oxidation. It

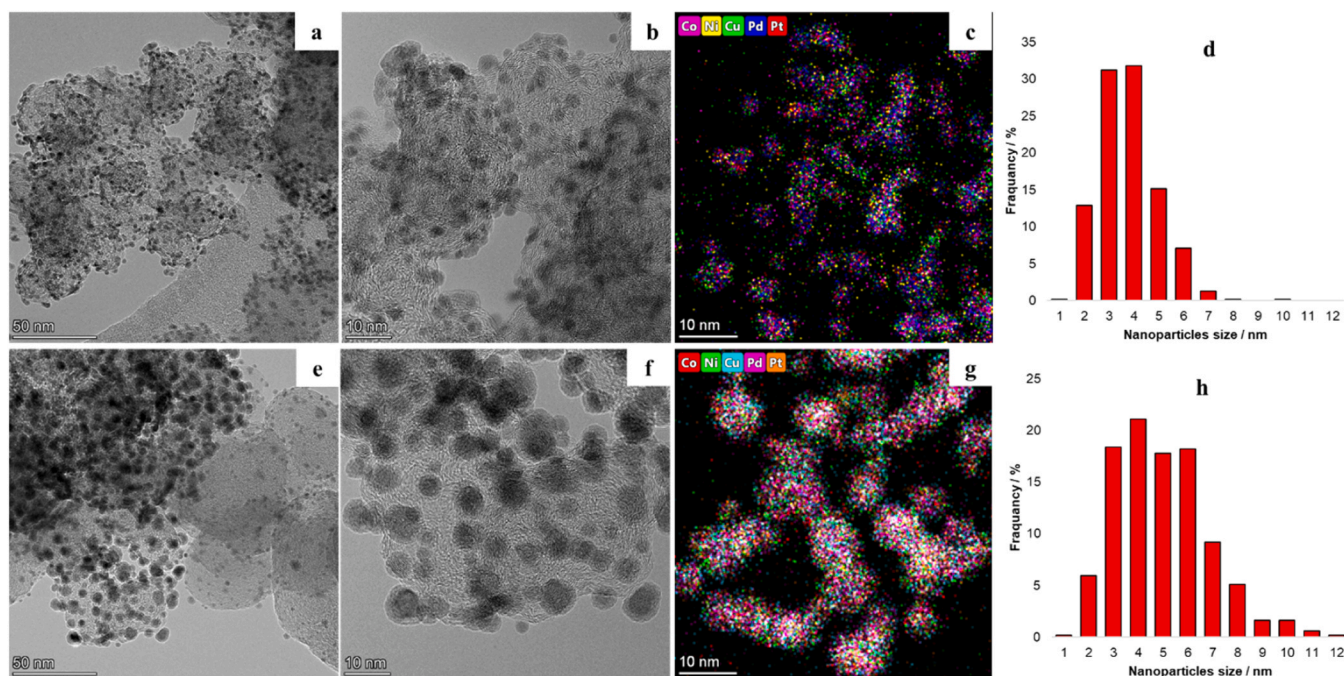


Fig. 2. Transmission electron microscopy images of the PtPdCuNiCo/C BG catalyst (a, b), EDX mapping of the BG catalyst (c), the BG-350 catalyst after heat treatment at 350 °C (e, f), EDX mapping of the BG-350 catalyst (g), and the corresponding particle size distribution histograms obtained from TEM image analysis (d, h).

also provides characteristic thermal parameters (e.g., onset, peak temperature, and range of intensive carbon combustion) and reveals how these are affected by the composition and microstructure of the deposited metal nanoparticles [56,57]. As shown (Fig. S1a) that after heat treatment at 600 °C, the mass fraction of the residue after combustion increased by approximately 4% compared to the material before treatment, which may be related to partial oxidation of the carbon support during the treatment. Analysis of the DSC curves reveals that after heat treatment, the maximum of carbon oxidation shifts from 437 °C for the BG material to 454 °C and 504 °C for the BG-350 and BG-600 materials, respectively (Fig. S1b). The combustion of carbon at higher temperatures after heat treatment may be associated both with partial oxidation of the most active carbon during the treatment and with the coarsening of metal particles, which leads to deteriorated catalytic performance for carbon oxidation.

The influence of heat treatment on the surface chemical composition of the obtained catalysts was studied by XPS (Fig. S8). The corresponding Pt4f, Pd3d, Cu2p, Ni2p, and Co2p spectra are shown in Fig. S5. The states with the Pt4f_{7/2} binding energy at ~71.2 eV and the Pd3d_{5/2} binding energy at ~335.5 eV are attributed to metallic Pt [58,59] and metallic Pd [60], respectively (Table 1).

A slightly higher value of the Pd3d_{5/2} binding energy than in bulk metallic palladium may arise from the interaction of small Pd clusters with the conductive carbon support, since for small clusters there is a possibility of binding energy change due to interatomic charge transfer to the support. In the spectrum of the BG sample, an additional doublet with the Pd3d_{5/2} binding energy of 337.1 eV is observed, as well as several small satellite peaks. This Pd3d doublet can be attributed to palladium in the Pd²⁺ state (fraction about 32%). Thus, in the "as-prepared" BG material, palladium is in a partially oxidized state and is reduced to metallic palladium already upon treatment at 350 °C, whereas platinum is predominantly in the metallic state.

In the Cu2p_{3/2} spectra of the BG, BG-350, and BG-600 samples, a narrow peak with a binding energy of 932.1 eV is observed (Fig. S8), corresponding to copper in the metallic state, along with a peak near ~934.2 eV, as well as several satellite peaks, corresponding to copper in the Cu²⁺ state [61,62]. Based on the obtained XPS data, it can be

Table 1

Binding energies of Pt4f_{7/2}, Pd3d_{5/2}, Cu2p_{3/2}, Ni2p_{3/2}, Co2p_{3/2}, C1s, and O1s for the BG catalyst with composition Pt₁₉Pd₁₆Cu₂₂Ni₂₂Co₂₁ synthesized by the borohydride method, before and after heat treatment at 350 and 600 °C.

Sample	C1s, eV	O1s, eV	Pt4f _{7/2} , eV	Pd3d _{5/2} , eV (%)	Cu2p _{3/2} , eV (%)	Ni2p _{3/2} , eV (%)	Co2p _{3/2} , eV (%)
BG	284.5	531.6 533.7	71.2	335.5 (68) 337.1 (32)	932.1 (33) 934.4 (67)	856.1	781.2
BG-350	284.5	530.0 531.8 533.7	71.2	335.5	932.1 (51) 934.2 (49)	854.5 (38) 856.1 (62)	779.6 780.6
BG-600	284.5	529.9 531.2 532.6 534.1	71.3	335.5	932.1 (70) 934.4 (30)	852.9 (35) 856.1 (65)	778.5 (30) 781.5 (70)

assumed that copper in the BG, BG-350, and BG-600 samples is in mixed Cu²⁺ and Cu⁰ states, with the fraction of metallic copper increasing with the treatment temperature (Table 1).

In the spectrum of the BG sample, a peak is observed with a Ni2p_{3/2} binding energy of 856.1 eV, accompanied by additional satellites shifted from the Ni2p_{3/2} peak by ~6 and ~10 eV (Fig. S8). This Ni2p_{3/2} binding energy value and the presence of these satellites, determined by multi-electron processes, are characteristic of Ni²⁺ in nickel hydroxide, Ni(OH)₂ [63]. A peak at 856.1 eV is observed in the Ni2p spectrum of all samples, from which it can be assumed that nickel is present as nickel hydroxide. In the spectrum of the BG-350 sample, an additional peak is observed with a Ni2p_{3/2} binding energy of 854.5 eV, characteristic of Ni²⁺ in NiO. In the spectrum of the BG-600 sample, an additional peak is observed with a Ni2p_{3/2} binding energy of 852.9 eV, corresponding to nickel in the metallic state. Thus, during heat treatment of the BG material, nickel partially (about 35%) transforms from the hydroxide form to the oxide form at 350 °C and to the metallic form at 600 °C.

In the Co2p spectrum of the BG sample, the Co2p_{3/2} peak at 781.2 eV

is observed with the corresponding satellites, which allows us to assign cobalt to Co^{2+} in the CoO structure (Fig. S8). For the BG-350 sample, there is an additional peak in the 779.2–779.6 eV region, which can be attributed to Co^{3+} [61]. In the BG-600 sample, cobalt is present in the Co^{2+} state and in the metallic Co^0 state (about 30%). Thus, in the studied samples, cobalt is predominantly in the Co^{2+} state; the BG-350 sample also contains cobalt atoms in the Co^{3+} state, and the BG-600 sample contains Co^0 .

The O1s spectra of the samples are described by two peaks in the 531.9–532.4 and 533.3–533.9 eV regions corresponding to oxygen in hydroxides and oxygen with an O–C bond, and to oxygen with a C=O bond and/or in adsorbed water, respectively [60]. In addition to these two peaks, the spectrum of the BG-350 sample contains an additional peak in the 530.0–530.5 eV region, which is usually attributed to lattice

oxygen in metal oxides (CoO , CuO , etc.). In the BG-600 sample, this peak is also present, but with significantly lower intensity, indicating the presence of a small amount of metal oxides in the sample.

Thus, in the PtPdCuNiCo/C BG material obtained by the wet-synthesis method, Pt is in the metallic state, Pd is in both metallic and oxidized states, and Ni, Co, and Cu are only in oxide/hydroxide forms. Heat treatment of the BG material at 350 ° and 600 °C leads to a change in the surface composition due to the partial reduction of Pd, Ni, Co, and Cu from oxide forms to the metallic state. It is noted that for Ni, Co, and Cu, even at 600 °C, despite the reducing atmosphere, a complete transition of the metals to the M^0 oxidation state is not observed. This fact may be related both to non-optimal treatment conditions and to subsequent oxidation of the components in air. To clarify this issue, additional *pseudo in situ* XPS studies of the BG material were conducted in an

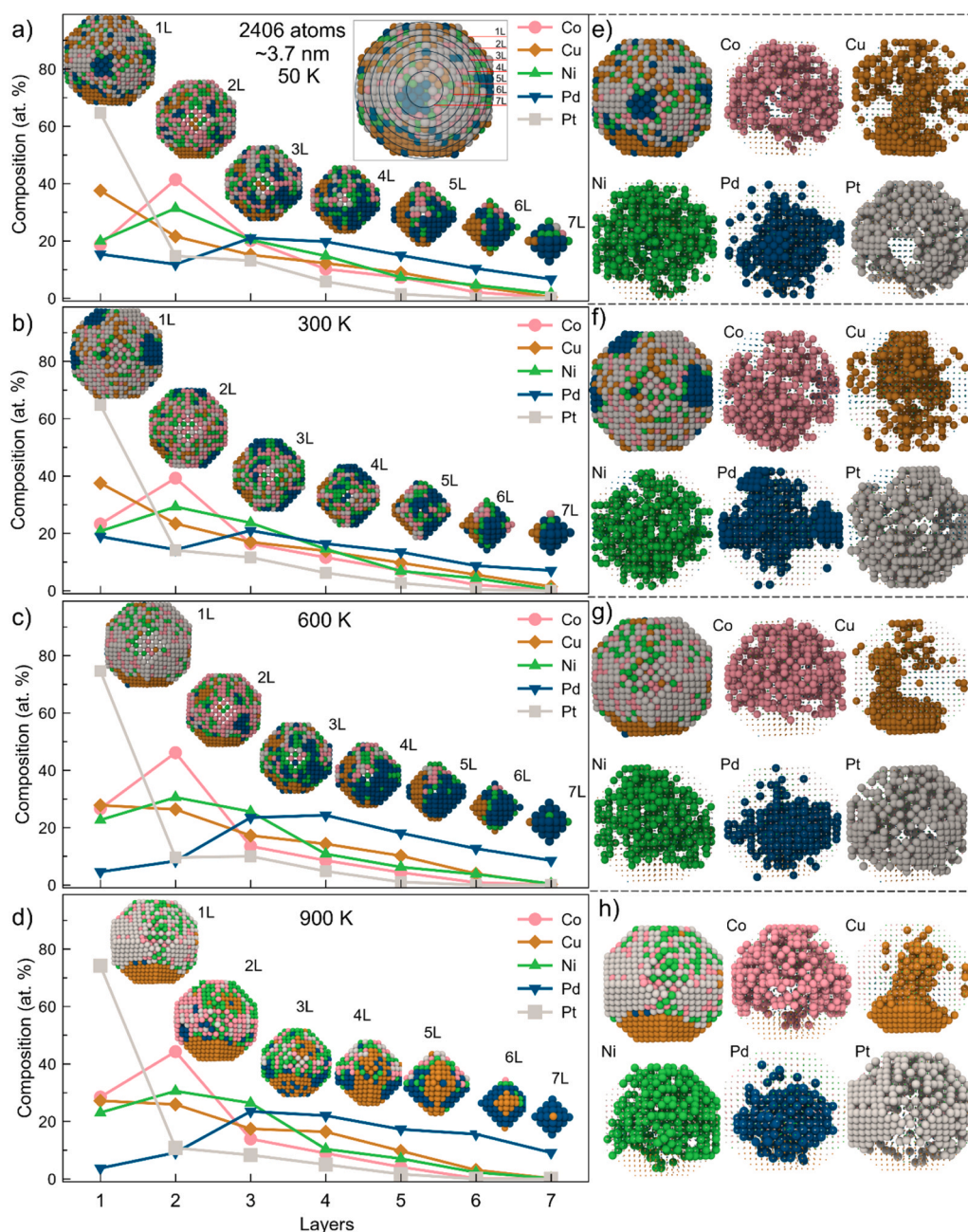


Fig. 3. Layer-by-layer distribution of the principal components of HEA (CoCuNiPdPt) nanoparticles with sizes of 4.2 nm at temperature a) 50 K, b) 300 K, c) 600 K, d) 900 K depending on the depth of the layer. L is the layer index counted from the nanoparticle's outer surface inward. Insets show the atomic structure of the nanoparticles. e-h) Spatial distribution of atomic species across the nanoparticle.

atmosphere of 5% H₂/Ar at 350 °C. It was shown that after treatment of the catalyst, palladium, copper, and nickel were completely reduced to the metallic state, while cobalt was reduced only partially and exists in the Co²⁺, Co³⁺, and Co⁰ states (Table S2, Fig. S9), in contrast to conventional treatment in a tube furnace followed by extraction and storage in air. Thus, only cobalt is not completely reduced at 350 °C in a 5% H₂/Ar atmosphere, while copper and nickel are reduced to the metallic state but are subsequently partially oxidized during catalyst storage in air.

Since the photoelectron escape depth is 3–4.6 nm, in this case, it is impossible to assess the surface composition by this method; therefore, additional approaches are needed to predict the surface composition of five-component nanoparticles.

3.3. Hybrid MD/MC simulations of the PtPdCuNiCo/C catalyst structure at different temperatures

To elucidate the equilibrium surface structure of the PtPdCuNiCo HEA nanoparticles, we analyzed the layer-by-layer elemental distributions obtained from the hybrid MD/MC simulations at four distinct temperatures: 50 K, 300 K, 600 K and 900 K. The resulting composition profiles across 7 at. layers from the surface (Layer 1) to the particle core are presented in Fig. 3. It is important to note that Fig. 3 shows the distribution of each element across the atomic layers, where the values represent the percentage of the total number of atoms of a given element in the entire nanoparticle that reside in a specific layer. For example, a value of 64% for Pt in Layer 1 indicates that 64% of all Pt atoms in the 2406-atom nanoparticle are located in the outermost surface layer. The sum of percentages for each element across all layers equals 100%. The data reveal a pronounced and systematic surface segregation that is strongly dependent on the annealing temperature.

For all simulated temperatures, a clear enrichment of Pt in the outermost surface layer (Layer 1) is observed. At 50 K and 300 K, approximately 64–65% of all Pt atoms in the nanoparticle are located in the first atomic layer. At 600 K, this surface enrichment of Pt intensifies further, with nearly 75% of all Pt atoms segregating to the outermost layer. This finding is in excellent agreement with the trends predicted by earlier surface energy calculations for pure metals [45], according to which Pt possesses one of the lowest surface energies among the constituent elements, making its segregation to the surface thermodynamically favorable for minimize the overall energy of the nanoparticle.

Conversely, a striking anti-segregation behavior is observed for Pd. While Pd is evenly distributed throughout the nanoparticle at the equiatomic composition, it strongly migrates to the inner core regions. At 50 K and 300 K, the majority of Pd atoms (over 80–85%) are located in Layers 3–7, with the deepest layers (Layers 6–7) containing nearly all of their atomic sites occupied by Pd. At 600 K, this effect becomes even more pronounced, with over 95% of all Pd atoms residing below the second layer, and the surface layer containing only a minimal fraction (approximately 4–5%) of the total Pd atoms. This behavior can be rationalized by Pd's higher surface energy compared to Pt and its different atomic size, which makes subsurface and core positions energetically more favorable.

The segregation patterns of the other 3d transition metals (Co, Cu, Ni) show intermediate behavior. Cu exhibits a moderate tendency for surface enrichment, with approximately 35–38% of all Cu atoms located in the first two layers at 50 K and 300 K. However, at 600 K, the surface Cu fraction decreases from ~38% to ~28%, suggesting that at higher temperatures, the driving force for Pt segregation dominates. Ni shows a relatively homogeneous distribution with a slight preference for subsurface layers (Layers 2–3), where 25–30% of all Ni atoms are located. Co is noticeably depleted from the surface layer at low temperatures, with the majority of Co atoms concentrated in the second and third layers, forming a Co-rich subsurface region. An increase in temperature to 900 K does not lead to a noticeable change in the atomic distribution within the nanoparticles compared to 600 K, while the segregation of copper atoms becomes even more pronounced (Fig. 3h).

This temperature-dependent behavior reflects the competition between enthalpic driving forces (favoring specific elemental arrangements) and entropic contributions (favoring randomization). At each temperature, the system reaches its thermodynamic equilibrium, and the equilibrium configuration varies with temperature, revealing the most stable configuration, a Pt-skin surface enriched with the majority of Pt atoms over a Pd-enriched core. This aligns with the conceptual framework proposed by Kristoffersen and Rossmeisl [46], according to which the width of the energy distribution of random atomic configurations (i.e. the standard deviation of ensemble energies) determines how strongly the system tends to adopt ordered structures. A wider distribution indicates a greater driving force for local ordering, and higher temperatures can actually enhance segregation towards the most stable configurations if the energy landscape has deep minima [46].

These findings have profound implications for catalysis. The formation of a Pt-enriched surface with minimal exposure of other metals (Pd, Cu, Co, Ni) at elevated temperatures (600 and 900 K) suggests that synthesis or annealing protocols at moderate temperatures could produce nanoparticles with a high density of catalytically active Pt sites while burying less active or potentially poisoning elements, including Pd, Cu, Co and Ni, in the core. According to the Sabatier principle for reactions like the oxygen reduction reaction (ORR), Pt is known to have a near-optimal binding energy for key intermediates. Therefore, the Pt-rich surface identified in our simulations provides a structural basis for understanding and predicting the high catalytic activity of PtPdCuNiCo HEA nanoparticles. Furthermore, the temperature-dependent tunability of the surface composition offers a potential lever for optimizing the balance between activity and stability by controlling the density of active Pt sites and the local coordination environment of neighboring elements [64–66].

Although the XPS results do not directly correspond to the nanoparticle surface composition predicted by MD/MC simulations, since the calculations indicate platinum enrichment in the surface layer while the XPS probing depth is 3–4.6 nm, such a comparison is nevertheless of interest. It was shown that the compositions of the PtPdCuNiCo/C material obtained by different methods correlate. For example, for the BG material, the XPS composition was Pt₃₂Pd₁₄Cu₂₀Ni₂₀Co₁₅, indicating platinum enrichment of the nanoparticle surface. However, according to MD/MC simulations, the surface composition at 300 K was Pt₃₉Pd₁₁Cu₂₃Ni₁₃Co₁₄, demonstrating even greater platinum enrichment and a reduced palladium content on the surface. During heat treatment, according to XPS data, the composition of the material remains virtually unchanged, while according to MD/MC simulations, the platinum content on the surface increases from 39 to 48 at. percent with an increase in temperature from 300 K to 600 K and does not change upon further heating to 900 K. Such differences may be related to the restructuring of the outer layers of the nanoparticles, which cannot be observed by XPS due to the relatively large photoelectron escape depth.

3.4. Activity and durability of the PtPdCuNiCo/C catalyst depending on heat treatment conditions

Determination of the electrochemically active surface area (ECSA) of the catalyst by cyclic voltammetry (CV) (Fig. 4) showed that heat treatment of the PtPdCuNiCo/C BG material leads to a decrease in the ECSA from 70 m²/g(PGM) to 55 and 31 m²/g(PGM) at treatment temperatures of 350 and 600 °C, respectively (Table 2). Measurement of the ECSA values from the oxidation of a chemisorbed CO monolayer shows a similar trend, with the ECSA values obtained by different methods agreeing within the error margin. It is noted that for the BG material, the CV shows two CO oxidation peaks: a main peak at a potential of 0.86 V and an additional, less intense peak at 0.77 V. It should be noted that for Pt/C, the CO oxidation peak maximum is typically around 0.8–0.82 V [67], and the Pd oxidation peak is around 0.91–0.92 V. The presence of an additional CO peak may be associated with the inhomogeneity of the nanoparticle composition or size [68]. After heat treatment at 350 °C,

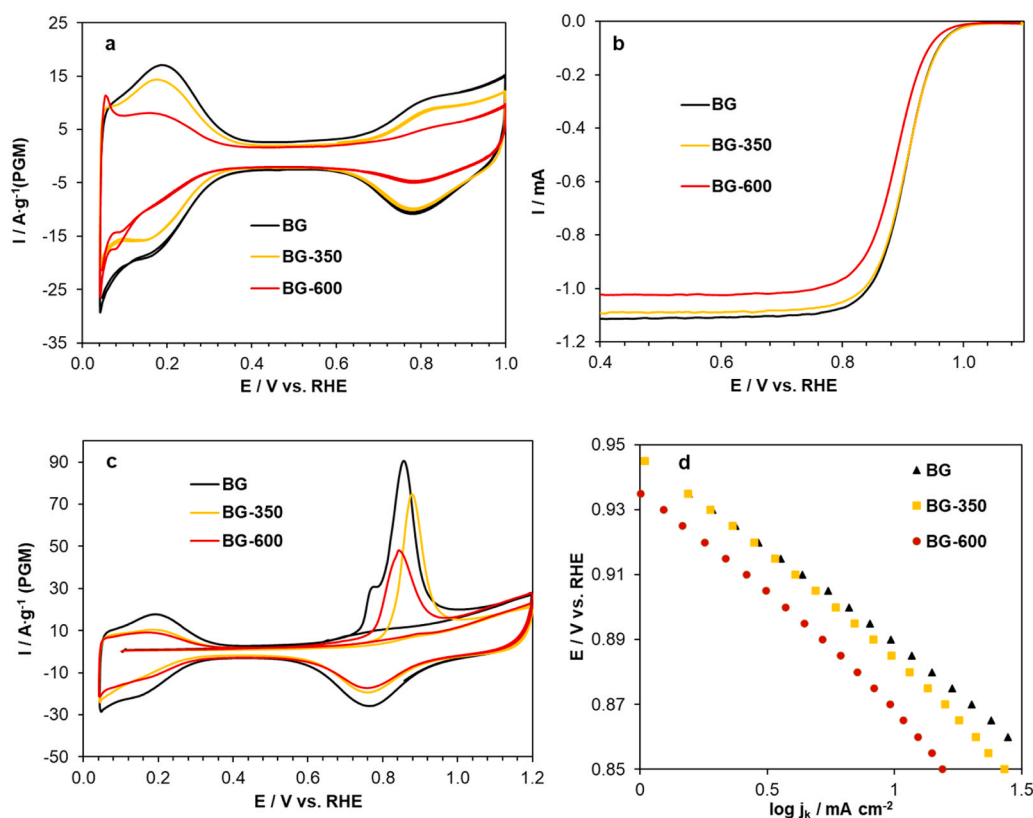


Fig. 4. Cyclic voltammograms (a), linear sweep voltammograms at 1600 rpm (b), and CO-stripping voltammograms (c), and Tafel plots (d) for the PtPdCuNiCo/C BG catalyst synthesized by the borohydride method before and after heat treatment at different temperatures. Electrolyte 0.1 M HClO₄.

Table 2

Characteristics of the PtPdCuNiCo/C catalysts and the commercial Pt/C analogue.

Sample	ω M, %	ω (Pt+Pd), %	ECSA, m ² /g(PGM)		I_m , A/g(PGM) E = 0.9 V	I_s , A/m ² (PGM) E = 0.9 V
			H _{ads}	CO		
BG	36.7	21.2	72	75	303	4.3
BG-350	36.7	21.2	55	50	275	5.0
BG-600	36.7	21.2	31	36	170	5.5
JM 20	20.0	20.0	80	82	254	3.2
JM 40	40.0	40.0	59	61	211	3.6

only one peak with a maximum at about 0.87 V is observed in the CV, and after treatment at 600 °C, the maximum shifts to a less positive potential region, down to 0.84 V. This indirectly indicates the preferential segregation of platinum on the nanoparticle surface, since the CO oxidation maximum is noticeably closer to that of Pt/C than to Pd/C. This fact correlates well with the platinum enrichment of the HEA nanoparticle surface composition according to the results from the hybrid MD/MC simulations. The change in the shape and position of the CO oxidation maxima after heat treatment may indicate a change in the surface composition, although an increase in particle size can also affect the position of the CO oxidation maximum. The regular decrease in the ECSA value is explained by the coarsening of metal nanoparticles during sample heating, which is more pronounced with increasing temperature.

It should be noted that the ECSA value for the PtPdCuNiCo/C BG material is higher than that for the commercial analog with a Pt loading of 40% (JM40), but lower than that for JM20 with a loading of 20%. The use of two commercial materials with different loadings is due to the fact that the PGM content in the as-prepared BG material is about 20%, while the total metal content approaches 40%. Thus, the use of different commercial materials allows for a more comprehensive comparison,

although the most important comparison is with the JM20 catalyst, which has similar PGM content.

The catalytic activity of the materials measured at 0.90 V vs. RHE was normalized both to the mass of PGM (A/g(PGM)) and to the ECSA (A/m²(PGM)). At the same time, the catalyst activity at a potential of 0.9 V after heat treatment also decreases from 303 A/g(PGM) for the BG catalyst to 275 and 170 A/g(PGM) after treatment at 350 and 600 °C, respectively. It is noted that the specific activity calculated per surface area increases with increasing treatment temperature from 4.3 to 5.5 A/m²(PGM), indicating an increase in the catalyst surface activity due to both the effect of an increased degree of alloying (decrease in the crystal lattice parameter) and a change in the surface state, as confirmed by the XPS results.

HEA catalysts are frequently investigated not only in acidic but also in alkaline environments [29,69,70], as their catalytic activity is highly pH-dependent [71]. To evaluate the catalytic activity of the obtained materials toward the ORR, measurements were also conducted in 0.1 M KOH using a rotating disk electrode (Fig. S10). The PtPdCuNiCo/C BG catalyst delivered the highest MA of 182 A/g(PGM) at 0.9 V, followed by BG-350 (168 A/g(PGM)) and BG-600 (146 A/g(PGM)), with the latter being comparable to the commercial Pt/C catalyst JM40 (122 A/g(PGM)) under identical conditions. These results confirm that all three synthesized materials outperform the commercial benchmark in alkaline media, with BG delivering a ~1.5-fold higher mass activity than JM40, which may be associated with the influence of residual oxide species of Co and Ni, similar to those observed by Zhang [72].

Thus, with increasing treatment temperature, on one hand, the ECSA value decreases due to the growth of nanoparticle size, and on the other hand, the specific activity of the catalysts increases due to the surface segregation of platinum atoms.

Based on a comparative assessment of the stability of the obtained PtPdCuNiCo/C catalyst BG-350 and a commercial counterpart, the HEA material was shown to exhibit significantly higher stability (Fig. S11).

Specifically, the degradation of the PtPdCuNiCo/C catalyst in terms of ECSA value after 10,000 cycles was 22%, whereas for the commercial Pt/C material it was 40%. Even more pronounced differences were observed in the activity at 0.9 V: while the activity of the commercial counterpart decreased by 36% to 163 A/g(PGM), the activity of the PtPdCuNiCo/C catalyst, on the contrary, increased to 668 A/g(PGM). Thus, after stress testing, the investigated catalysts show comparable ECSA values but substantially different activities, indicating a fourfold higher surface-specific activity of the PtPdCuNiCo/C catalyst compared to the commercial benchmark. This observation may be related to changes in the surface composition of the HEA catalyst [73]; a similar result was reported in [74]. The obtained result demonstrates the promise of using the synthesized catalysts due to their enhanced stability and calls for further detailed investigation of the changes in composition and structure of the catalyst during stress testing, which is beyond the scope of the present work.

4. Conclusions

A PtPdCuNiCo/C catalyst was obtained by wet-synthesis using sodium tetrahydroborate to study the evolution of the structure of five-component nanoparticles during heat treatment. Based on the results of *in situ* XRD during heating in a reducing atmosphere, not only were the processes of nanoparticle coarsening and changes in the crystal lattice parameter during atomic interdiffusion shown, but also phase segregation at temperatures above 400 °C. TEM data confirmed the formation of five-component PtPdCuNiCo nanoparticles, which coarsen upon heat treatment. XPS studies of the obtained PtPdCuNiCo/C catalysts showed that in the initial material before heat treatment, Pt is in the metallic state, Pd is in both metallic and oxidized states, and Ni, Co, and Cu are present only in oxide/hydroxide forms. Heat treatment of the PtPdCuNiCo/C material at 350 and 600 °C leads to a change in the surface composition due to the partial reduction of Pd, Ni, Co, and Cu from oxide forms to the metallic state. According to the results of a *pseudo in situ* XPS study of the PtPdCuNiCo/C BG material in an atmosphere of 5% H₂/Ar at 350 °C, it was shown that after catalyst treatment, palladium, copper, and nickel are completely reduced to the metallic state, while cobalt is reduced only partially and remains in a partially oxidized state. Thus, only cobalt is not completely reduced at 350 °C in a 5% H₂/Ar atmosphere, while copper and nickel are reduced to the metallic state but are subsequently partially oxidized during catalyst storage in air. Calculation of the surface composition of HEA nanoparticles based on the results from the hybrid MD/MC simulations shows a significant enrichment of the nanoparticle surface with platinum, which correlates well with the results of the electrochemical behavior of the catalysts. Heat treatment leads both to the enrichment of the surface with platinum and an increase in the specific surface activity, and to the coarsening of nanoparticles and a decrease in the ECSA value. These two opposing trends determine the existence of an optimal treatment temperature. Under the conditions of this study, heat treatment at 350 °C strikes a balance between enhanced activity and avoidance of significant phase separation. The excellent stability of the PtPdCuNiCo/C catalyst after heat treatment at 350 °C has been demonstrated, significantly surpassing that of the commercial Pt/C analogue. Specifically, the ORR activity of the HEA catalyst after stress testing is four times higher than that of the Pt/C benchmark. This result highlights the practical significance of this work and the promising potential of the obtained catalysts for use in low-temperature fuel cells. Furthermore, it underscores the need for a more detailed investigation into the reasons for the enhanced stability of HEA catalysts, which will be conducted in future studies.

CRedit authorship contribution statement

Alexander G. Kvashnin: Visualization, Data curation. **Belenov Sergey V:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Ilya V. Chepkasov:**

Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Anna M. Kremneva:** Writing – original draft, Visualization, Data curation. **Valeria P. Konovalova:** Writing – original draft, Visualization, Formal analysis, Data curation. **Viktor S. Baidyshev:** Visualization, Software, Data curation. **Evgeniy Yu. Gerasimov:** Visualization, Data curation. **Dmitry K. Mauer:** Visualization, Formal analysis, Data curation. **Anna A. Gavrilova:** Visualization, Investigation, Formal analysis, Data curation. **Alina K. Nevelskaya:** Visualization, Methodology, Investigation, Formal analysis, Data curation. **Egor R. Beskopylny:** Visualization, Software, Formal analysis, Data curation.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2026.190381](https://doi.org/10.1016/j.jallcom.2026.190381).

Data availability

All data are presented in the main text and supplementary materials.

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